

PMID 37931033 — Figure 1 Data Tables (as analyzed)

Parkman et al., Mol Cancer Ther 2024, Figure 1 (Panels A–D) and Supplementary Figures S3–S4. Read directly from the primary figures. See evidence_training_examples.md Rows 2–4 for the full graded/curated version.

Table 1 — Figure 1A: Proliferation (confluence assay), all 6 cell lines × 8 conditions

Condition	A375	HT144	SK-MEL28	WM793 (Pten-/-)	MTG001	MTG004
siAKT1 alone	No effect	No effect	No effect	No effect	No effect	No effect
siAKT2 alone	No effect	No effect	No effect	No effect	No effect	No effect
siAKT3 alone	No effect	No effect	No effect	No effect	No effect	No effect
siAKT1+2 (AKT3 intact)	No effect	No effect	No effect	No effect	No effect	No effect
siAKT2+3 (AKT1 intact)	No effect	No effect	No effect	No effect	No effect	No effect
siAKT1+3 (AKT2 intact)	No effect	Effect	Effect	No effect	No effect	Effect
siAKT1+2+3 (all silenced)	Effect	Effect	Effect	Effect	Effect	Effect

Pattern: AKT1 alone and AKT3 alone are each sufficient in all 6 lines. AKT2 alone is sufficient in only 3 of 6 lines (A375, WM793, MTG001) and insufficient in the other 3 (HT144, SK-MEL28, MTG004). Triple knockdown is the only condition with zero exceptions.

Table 2 — Figure 1D: Cell death (Cytotox Red assay), A375 only

Condition	Cell death observed?	Notes
Staurosporine (positive control)	Yes — gradual increase	Pan-kinase inhibitor, established cytotoxic control
siCtrl (negative control)	No	Baseline
siAKT1+2+3 (triple knockdown)	Yes — dramatic increase at ~22h	Comparable magnitude to staurosporine
siAKT1+2+3 + WT mouse Akt1 (rescue)	No — rescued	Death abrogated by AKT1 reintroduction

No pairwise condition (e.g. siAKT1+3) was tested in the death assay — only proliferation (Table 1) was measured for pairwise combinations.

Table 3 — Verification coverage (Figure 1B/C + Supplementary Figure S3)

Cell line	Knockdown confirmed	Method	Cross-reactivity checked (all 3 paralogs)	Activity readout
A375	Yes	Western blot	Partial (siAKT1, siAKT2 only)	P-AKT(S473), Fig 1C
WM793	Yes	Western blot	No (siAKT3 blot shows AKT3 band only)	P-PRAS40
HT144	Yes	Western blot	Yes (all 3 conditions × all 3 antibodies)	P-PRAS40
MTG001	Yes	qPCR	N/A (qPCR, not blot)	Not tested
MTG004	Yes	qPCR	N/A (qPCR, not blot)	Not tested
SK-MEL28	Not found	—	—	—

Table 4 — Supplementary Figure S4: mouse-specificity control (YUMM1.1, mouse cell line)

Condition	Effect on mouse AKT expression/activity?
Human-specific siAKT1, siAKT2, siAKT3, and all combinations (including triple)	No effect — confirms siRNAs are human-specific, don't cross-react with mouse AKT (P-AKT/P-PRAS40 blot unchanged vs. siCtrl)

Validates that the mouse AKT1 construct used in the Figure 1C-D rescue is genuinely resistant to the human-specific siRNAs, not itself being silenced.

Key gaps/flags established during analysis (see evidence_training_examples.md for full curator notes): Rescue-construct data exists only for AKT1, despite the paper's text stating all three paralogs were tested and concluding "any one paralog...can compensate for the other two." Endogenous pairwise-knockdown data (Table 1) independently supports AKT1 and AKT3 standalone sufficiency in all lines, and shows AKT2 standalone sufficiency is cell-line-dependent. Cell death (Table 2) was only tested for the triple-knockdown condition — reduced proliferation under partial knockdown was never shown to translate to actual cell death in any line.